Radix-14 rotation-cache transfer and panel: diagnostic report

Question. Do training-hot digit seeds transfer to an unseen scalar stream, and does caching two-x rotation data in ten digit rows improve the seven-row reference under the frozen 32 MiB cap?

Scope and inputs. This report covers two linked but distinct results for SEC 2 secp256k1 over `p = 2^256 - 2^32 - 977`, subgroup order `n = FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364141`. The transfer screen uses 20,000 SHAKE256-derived training scalars and a separate 12,288-scalar evaluation fixture; their derivation domains differ. The training and evaluation histogram archives are SHA-256 `5bfc62f10c60bcf0bc47ef68db8c1cc99561037fad8218d5` and `34aacf4063e4f5716f5938dd513d572a57ea31e59cd87ed3de6218718fc6f93d`. The held-out fixture is SHA-256 `a1c4964f174e36ea26f6a1897128f651a2ce7251e4bf4ffd532ff09221e721d3`.

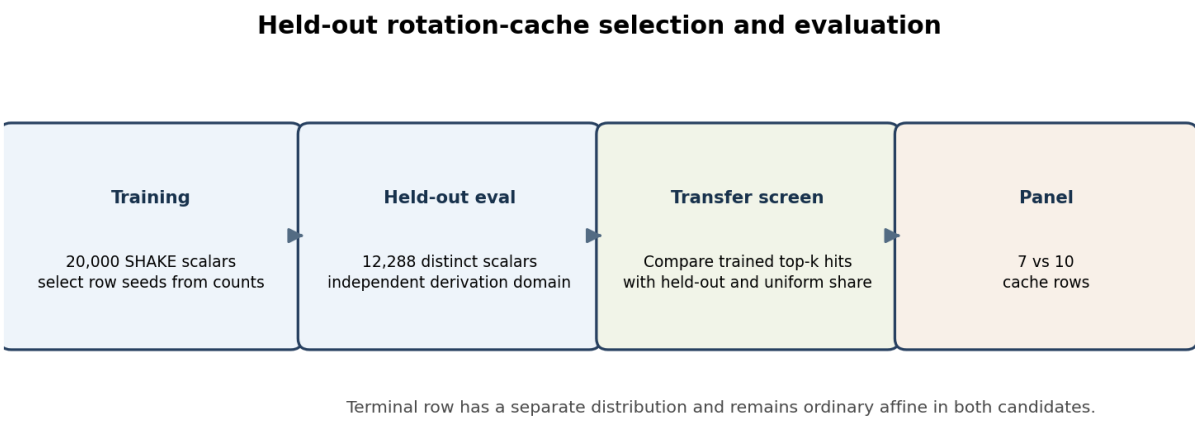


Figure 1: Training, held-out evaluation, transfer screen, and cache panel sequence

Transfer method and finding. For each of the fourteen digit rows, the analysis sorts seeds by training count and measures their share of training and held-out selections at `k=100, 1,000, 5,000, 10,000`, against the uniform share `k/22,756`. The reanalyzer was rerun from decompressed committed histograms and reproduced `rotation-seed-transfer-audit.json` byte-for-byte. Mean training/evaluation Pearson correlation across those fourteen rows is `0.003705`. For top 1,000 seeds per row, mean training coverage is `16.6972%`, held-out coverage is `4.4789%`, and uniform coverage is `4.3944%`. Training-hot seeds therefore show no useful transfer on the held-out rows. The terminal row is analyzed separately because its distribution differs; it remains ordinary affine in both candidates.

Cache panel and measurement boundary. The ten-row overlay selects rows `0,1,3,4,6,7,9,10,12,13` from the training-based rule without consulting held-out scalars. The reference stores two-x values in seven even rows. The frozen serial Tier 1 panel used 16 alternating pairs, 96 arms, 48 pairs, 12,288 checked outputs per arm, a fixed 33,554,432-byte cap, and one RunPod instance (`thyvmdqmzsz5ik`, CPU 40). It measures 96 consecutive 128-scalar batches after fixture loading and table construction, with output checking outside the timed interval; preparation and retained bytes are separate.

The independent panel auditor was rerun and reproduced `tier1-cache10-cpu40-audit.json` byte-for-byte. Native provenance archive SHA-256 is `0320bddb984f3f3afc958d0d2e2467779596ebc3516f612f07ded96cf25ff6da`; raw panel archive SHA-256 is `0f68e3e2c1c54ba00013b803b6a135b484b9c3e1742286650952603f0e087f47`; job ID is `e0d10d9a14e4406684212690eb680518`. The reference averaged 67.807406 ms and retained 31,133,444 bytes; the candidate averaged 65.714195 ms and retained 33,318,020 bytes. The descriptive paired median reference/candidate ratio is `1.028614`, with bootstrap interval `[1.014163,1.046736]`, but two A/A controls were outside the predeclared `[0.90,1.11]` band (`1.203446`, `1.201942`). Under the frozen gate the qualified ratio is null and status is `unknown_after_tier_gate`.

Decision and limits. The held-out screen rejects selecting hot seeds from the 20,000-scalar profile. The ten-row panel does not establish a performance winner: the candidate uses about 2.19 MB more table memory, and failed A/A controls invalidate the timing aggregate. The exact replay and retained-memory counts pass, but this is a batch result on one observed cloud instance; physical-host isolation is unobserved. A new timing decision needs a newly frozen environment and manifest. The editable figure source is `rotation_cache_report_2026-10-19.py`; SVG/PNG outputs and a PDF accompany this report.

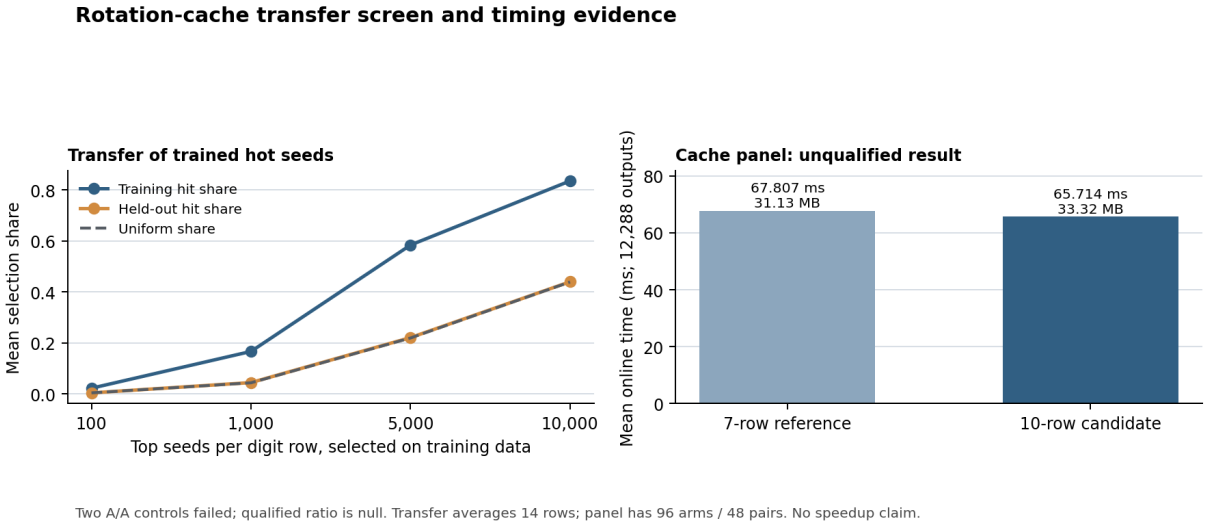


Figure 2: Held-out hot-seed transfer and unqualified seven-row versus ten-row cache timings

Open checks and graph status. The fixed row schedule is not shown to be globally memory-optimal. The replay does not prove constant-time behavior or formal correctness of the implementation. No new curve, endomorphism, isogeny, candidate route, or winner is proposed. Canonical curve/isogeny graph files are absent from this PR's experiment tree, so this report makes no claim that main-branch graph files were checked.